

Solutions

Numbering follows the lecture notes. For TYPE C exercises the idea is boxed; the rest are routine once the idea is in hand, which is what the types mean.

Lectures 1 and 2

§1.2–§1.4 Definition, existence, independence

1.8 A (a) $\mu \cdot \xi = \xi_1 - \xi_2$ and $\xi^\top \Sigma \xi = 2\xi_1^2 + 2\xi_1\xi_2 + \xi_2^2$, so $\varphi_X(\xi) = \exp(-i(\xi_1 - \xi_2) - \frac{1}{2}(2\xi_1^2 + 2\xi_1\xi_2 + \xi_2^2))$. Proposition 1.4 gives $\partial_1 \partial_2 \varphi_X(0) = -\Sigma_{12} - \mu_1 \mu_2 = -1 + 1 = 0 = -\mathbb{E}[X_1 X_2]$, so $\mathbb{E}[X_1 X_2] = 0$ — uncorrelated only after the means are subtracted.

(b) With $B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, Theorem 1.5 gives $BX \sim \mathcal{N}((0, 2), B\Sigma B^\top)$ and $B\Sigma B^\top = \begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix}$. So $X_1 + X_2 \sim \mathcal{N}(0, 5)$, $X_1 - X_2 \sim \mathcal{N}(2, 1)$, and their covariance is $1 \neq 0$: not independent.

(c) $\text{Cov}(X_1 + X_2, X_1 - X_2) = \Sigma_{11} - \Sigma_{22}$, and the pair is jointly Gaussian, so Theorem 1.16 makes independence equivalent to $\Sigma_{11} = \Sigma_{22}$ — equal variances, whatever the correlation. And $v \cdot X$ is constant iff $\text{Var}(v \cdot X) = v^\top \Sigma v = 0$, i.e. iff $v \in \ker \Sigma$, so such a v exists exactly when Σ is singular.

1.9 C

Apply the invariance to the **transform**, and rotate $(\xi_1, \xi_2, 0, \dots, 0)$ onto $(\sqrt{\xi_1^2 + \xi_2^2}, 0, \dots, 0)$. Independence makes φ_X a product; the rotation turns that product into Cauchy's functional equation in ξ^2 .

Independence gives $\varphi_X(\xi) = \prod_j \varphi_1(\xi_j)$, where φ_1 is the common transform of the coordinates. Invariance gives $\varphi_X(O^\top \xi) = \varphi_{OX}(\xi) = \varphi_X(\xi)$ for all O , so φ_X is a function of $|\xi|$ alone. Taking $O = \text{diag}(-1, 1, \dots, 1)$ gives $X_1 \stackrel{d}{=} -X_1$, so φ_1 is real and even.

Now take $\xi = (\xi_1, \xi_2, 0, \dots, 0)$ and an O carrying it to $(r, 0, \dots, 0)$ with $r = \sqrt{\xi_1^2 + \xi_2^2}$. Since $\varphi_1(0) = 1$, the two products agree:

$$\varphi_1(\xi_1) \varphi_1(\xi_2) = \varphi_1(\sqrt{\xi_1^2 + \xi_2^2}).$$

Set $F(u) = \varphi_1(\sqrt{u})$ for $u \geq 0$. Then $F(u)F(v) = F(u+v)$, $F(0) = 1$, and F is continuous. Also $F(u) = F(u/2)^2 \geq 0$; if $F(u_0) = 0$ for some u_0 then $F(u_0 2^{-k}) = 0$ for every k , contradicting $F(0) = 1$ and continuity. So $F > 0$, $\log F$ is continuous and additive, hence linear, and $F(u) = e^{-cu/2}$. Finally $|F| \leq 1$ forces $c \geq 0$. Therefore $\varphi_1(\xi) = e^{-c\xi^2/2}$ and $\varphi_X(\xi) = e^{-c|\xi|^2/2}$, i.e. $X \sim \mathcal{N}(0, cI_d)$.

No moment hypothesis was used anywhere; the variance appears at the end as the constant c . Note also where $d \geq 2$ entered: with one coordinate there is nothing to rotate, and $O(1) = \{\pm 1\}$ says only that X_1 is symmetric.

1.14 A $\hat{f}(\xi) = \int f(x) e^{-ix \cdot \xi} dx = \mathbb{E}[e^{-iZ \cdot \xi}]$ for $Z \sim \mathcal{N}(0, I_d)$, which factors into d copies of Lemma 1.10: $\hat{f}(\xi) = \prod_j e^{-\xi_j^2/2} = e^{-|\xi|^2/2}$.

For the inversion, each coordinate integral is $\int_{\mathbb{R}} e^{-\xi^2/2} e^{ix\xi} d\xi = \sqrt{2\pi} \mathbb{E}[e^{-i(-x)g}] = \sqrt{2\pi} e^{-x^2/2}$, again Lemma 1.10. Multiplying d of them and dividing by $(2\pi)^d$ gives $(2\pi)^{-d/2} e^{-|x|^2/2} = f(x)$. Since $\hat{f} = (2\pi)^{d/2} f$, the standard Gaussian density is an eigenfunction of the transform.

1.15 B (a) $Z = A^{-1}(X - \mu)$, so $p_X(x) = p_Z(A^{-1}(x - \mu)) |\det A|^{-1}$. Now $|A^{-1}(x - \mu)|^2 = (x - \mu)^\top A^{-\top} A^{-1} (x - \mu) = (x - \mu)^\top \Sigma^{-1} (x - \mu)$ because $\Sigma^{-1} = (AA^\top)^{-1} = A^{-\top} A^{-1}$, and $|\det A| = (\det \Sigma)^{1/2}$. Both quantities are functions of Σ alone, so the formula is independent of the square root chosen; concretely, any two differ by $A' = AO$ with O orthogonal, and $|\det A'| = |\det A|$.

(b) If Σ is singular pick $v \in \ker \Sigma$, $v \neq 0$. Then $\text{Var}(v \cdot X) = v^\top \Sigma v = 0$, so $v \cdot X = v \cdot \mu$ and X lies in the hyperplane $\{x : v \cdot x = v \cdot \mu\}$. Integrating any density over a hyperplane gives 0, so X has none.

1.18 B (a) $X + Y = (1 + \varepsilon)g$, which is 0 when $\varepsilon = -1$ and $2g$ when $\varepsilon = +1$: a half-and-half mixture of δ_0 and $\mathcal{N}(0, 4)$. It has an atom of mass $\frac{1}{2}$ at the origin. A Gaussian law is either a point mass or has a density, and this is neither.

(b) $sX + uY = (s + u\varepsilon)g$, a half-and-half mixture of $\mathcal{N}(0, (s+u)^2)$ and $\mathcal{N}(0, (s-u)^2)$, with transform $\frac{1}{2}e^{-(s+u)^2/2} + \frac{1}{2}e^{-(s-u)^2/2}$ at $\xi = (s, u)$. That is Gaussian only when $(s+u)^2 = (s-u)^2$, i.e. $su = 0$. So the hypothesis of Corollary 1.7 holds *only* on the two axes, which is exactly the statement that the marginals are Gaussian; off the axes it fails. The corollary is not contradicted — it never applied.

(c) No: (X, Y) lies on $\{y = x\} \cup \{y = -x\}$, and integrating a density over two lines gives 0.

§1.5* Entropy, and why the Gaussian maximizes it

1.19 A (a) $H(q) = -q \log q - (1-q) \log(1-q)$: symmetric about $q = \frac{1}{2}$, zero at the endpoints, maximal at $\frac{1}{2}$ with value $\log 2$. Uniform on n points: $\log n$.

(b) $\log a$; $1 - \log \lambda$; $\frac{1}{2} \log(2\pi e \sigma^2)$. Uniform on $[0, a]$ with $a < 1$ has $h = \log a < 0$. Discrete entropy cannot be negative because $-p \log p \geq 0$ for $p \in (0, 1]$; a *density* may exceed 1, which is exactly what breaks that.

(c) $p_Y(y) = |a|^{-1} p_X(y/a)$; substituting $x = y/a$, $h(Y) = -\int p_X(x) [\log p_X(x) - \log |a|] dx = h(X) + \log |a|$.

1.20 B $-D(p \| q) = \int p \log \frac{q}{p} \leq \log \int q = 0$ by Jensen, with equality iff q/p is constant, hence $\equiv 1$. Taking $q \equiv 1/n$ uniform on n points, $0 \leq D(p \| q) = -H(p) + \log n$, with equality only for $p = q$. Under $X \mapsto aX$ both densities gain the same factor $|a|^{-1}$ and the same substitution is made, so $D(p \| q)$ is unchanged.

1.21 B $\log q(x) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma - \frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)$ is quadratic in x , so $\int p \log q$ is a fixed linear functional of $\int p$, $\int x_i p$, $\int x_i x_j p$. Matching mean and covariance therefore gives $\int p \log q = \int q \log q = -h(q)$, and $0 \leq D(p \| q) = -h(p) + h(q)$. For $h(q)$: $\mathbb{E}[(X - \mu)^\top \Sigma^{-1} (X - \mu)] = \text{tr}(\Sigma^{-1} \Sigma) = d$, so $h(q) = \frac{1}{2} \log((2\pi e)^d \det \Sigma)$.

1.22 C

$(x, y) \mapsto (\frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}})$ is **orthogonal**: Jacobian 1, so it preserves joint differential entropy.

With $U = \frac{X+Y}{\sqrt{2}}$, $V = \frac{X-Y}{\sqrt{2}}$: $h(U, V) = h(X, Y) = 2h(X)$ by independence. Subadditivity $h(U, V) \leq h(U) + h(V)$ is Exercise 1.20 applied to the joint density against the product of its marginals. Symmetry gives $(X, Y) \stackrel{d}{=} (X, -Y)$, so $V \stackrel{d}{=} U$. Chaining, $2h(X) \leq 2h(U)$.

1.23 C

The Gaussian proof never used Gaussianity. It used that $\log q$ is an **affine combination of the constrained functions**, so $\int p \log q$ is fixed by the constraints alone.

$\log q = \sum_m \lambda_m T_m - \log Z$, so for any feasible p , $0 \leq D(p \| q) = -h(p) - \sum_m \lambda_m c_m + \log Z$ — and the last two terms do not depend on p . Taking $p = q$ evaluates them as $h(q)$; subtracting gives $h(p) \leq h(q)$, equality iff $p = q$. (No integrability of $h(p)$ need be assumed: $-\infty$ makes it trivial, $+\infty$ is excluded by $D(p \| q) \geq 0$.)

Instances: no constraint on $[0, a]$ gives q constant; $T_1(x) = x$ on $[0, \infty)$ gives the exponential; T 's $= x_i, x_i x_j$ give $\mathcal{N}(\mu, \Sigma)$.

Why the assumption is not free. On $S = \mathbb{R}$ with only the mean constrained, $e^{\lambda x}$ is never integrable, and correctly so: $\mathcal{N}(0, \sigma^2)$ is feasible for every σ with $h = \frac{1}{2} \log(2\pi e \sigma^2) \rightarrow \infty$. “ **q is a density**” is exactly what rules this out, and a student who assumes existence has assumed the content.

§2.3 The tower property

2.6 A Corollary 2.5 reads $\text{Var } X = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$ and the first term is non-negative. Equality holds exactly when $\text{Var}(X | Y) = 0$, i.e. when X is a function of Y . All the variance sits in the first term exactly when $\mathbb{E}[X | Y]$ is constant, as happens when $X \perp\!\!\!\perp Y$.

2.7 B Given J , X has mean ± 1 and variance σ^2 , so $\text{Var}(X | J) = \sigma^2$ identically and $\mathbb{E}[\text{Var}(X | J)] = \sigma^2$, while $\mathbb{E}[X | J] = \pm 1$ with $\text{Var}(\mathbb{E}[X | J]) = 1$. The total is $\sigma^2 + 1$. Directly: $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = \mathbb{E}[\mathbb{E}[X^2 | J]] = \mathbb{E}[\sigma^2 + 1] = \sigma^2 + 1$.

For the second part, W is a function of $X_1, \dots, X_n$, so conditioning on the X_i is the finer conditioning and Proposition 2.4 gives

$$\mathbb{E}[V | W] = \mathbb{E}[\mathbb{E}[V | X_1, \dots, X_n] | W] = \mathbb{E}[-W/n | W] = -W/n,$$

the last step because a function of W is its own conditional expectation given W . The property is the **tower property**.

§2.5 Bayesian linear models, §2.6* and §2.7*

2.12 A (a) $p(y | w) = (2\pi\sigma^2)^{-n/2} e^{-\|y - Hw\|^2 / 2\sigma^2}$, so $-\log p = \frac{1}{2\sigma^2} \|y - Hw\|^2 + \frac{n}{2} \log(2\pi\sigma^2)$ and the minimizer over w is the least-squares fit. Choosing squared error is asserting that the noise is Gaussian, homoscedastic and independent across observations; most people choosing it are not asserting anything, which is the point.

(b) The posterior density is $\propto \exp\left(-\frac{1}{2\sigma^2}\|y - Hw\|^2 - \frac{1}{2\tau^2}\|w\|^2\right)$, so its maximizer minimizes $\|y - Hw\|^2 + \lambda\|w\|^2$ with $\lambda = \sigma^2/\tau^2$. Large λ means noisy data against a confident prior, so heavy shrinkage; $\tau \rightarrow 0$ gives $\hat{w} = 0$ and $\tau \rightarrow \infty$ gives least squares.

2.13 B (w, y) is a linear image of (w, ε) , hence jointly Gaussian, with $\Sigma_{ww} = \tau^2 I_p$, $\Sigma_{wy} = \mathbb{E}[w(Hw + \sigma\varepsilon)^\top] = \tau^2 H^\top$, $\Sigma_{yy} = \tau^2 HH^\top + \sigma^2 I_n$. Theorem 2.10 then gives the two displays directly. $\text{Cov}(w | y)$ contains no y : the data changes your *estimate*, never your *uncertainty*. Operationally, a surprising y moves the posterior mean and leaves the error bars exactly where they were — special to the Gaussian model, and false in general.

2.14 B $\text{Cov}(X, Y) = \mathbb{E}[(Y^2 - 1)Y] = \mathbb{E}Y^3 = 0$, so the best linear predictor is 0 with error $\text{Var } X = \text{Var}(Y^2) = 2$. Here X is itself a function of Y , so $\mathbb{E}[X | Y] = Y^2 - 1 = X$ with error 0. What fails is **joint** Gaussianity of (X, Y) — X is not Gaussian at all — so Theorem 2.9 does not apply, and the conditional mean is not linear.

2.15 C

$H^\top(HH^\top + \lambda I_n) = (H^\top H + \lambda I_p)H^\top$ — one line of expansion. Multiply by the two inverses.

Both sides equal $H^\top HH^\top + \lambda H^\top$. Left-multiplying by $(H^\top H + \lambda I_p)^{-1}$ and right-multiplying by $(HH^\top + \lambda I_n)^{-1}$ gives the identity. The right-hand form inverts an $n \times n$ matrix, so the cost is $O(n^3)$; this is the kernel trick, and why Gaussian process regression is feasible.

As $\tau \rightarrow \infty$ we have $\lambda \rightarrow 0$, and in the form $H^\top(HH^\top + \lambda I_n)^{-1}y$ the limit is benign because $HH^\top$ is $n \times n$ and invertible: $\mathbb{E}[w | y] \rightarrow H^\top(HH^\top)^{-1}y$, the minimum-norm solution of $Hw = y$. In the other form the limit is $0 \cdot \infty$ — which is what makes this a C and not a computation.

2.17 A (X, ξ) is jointly Gaussian, so $Y = X + \sigma\xi \sim \mathcal{N}(0, (\tau^2 + \sigma^2)I_d)$ by Theorem 1.5 of Lectures 1–2. Hence $\log p_Y(y) = c - \frac{|y|^2}{2(\tau^2 + \sigma^2)}$ and $\nabla \log p_Y(y) = -\frac{y}{\tau^2 + \sigma^2}$, so Proposition 2.16 gives

$$\mathbb{E}[X | Y = y] = y - \frac{\sigma^2}{\tau^2 + \sigma^2} y = \frac{\tau^2}{\tau^2 + \sigma^2} y.$$

Theorem 2.9 gives $\Sigma_{XY}\Sigma_{YY}^{-1}y = \tau^2 I((\tau^2 + \sigma^2)I)^{-1}y$, the same. Only the second derivation used that X is Gaussian; in the first, Gaussianity of X entered nowhere except to make p_Y available in closed form.

2.18 B $p_Y(y) = \mathbb{E}[\phi_\sigma(y - X)]$. Every derivative of ϕ_σ is bounded and, on a compact set of y , bounded uniformly in X ; dominated convergence therefore justifies differentiating under the expectation to all orders, so $p_Y \in C^\infty$ with $\nabla p_Y(y) = \mathbb{E}[\nabla \phi_\sigma(y - X)]$.

For the second integration by parts, $\int \nabla f p_Y = -\int f \nabla p_Y$ requires $f p_Y \rightarrow 0$ at infinity. Taking f smooth with compact support makes the boundary term vanish outright, and that suffices: such f are dense in $L^2(p_Y)$, so orthogonality against them is orthogonality against all competitors in Lemma 2.3(ii). *Both steps are free because Y is a convolution with a Gaussian; an arbitrary Y need not have a density, let alone a differentiable one.*

2.19 C

Tweedie's formula **is** the dictionary. It is an explicit pointwise bijection between the optimal denoiser and the score, so producing either produces the other.

By Definition 2.2 the least-squares minimizer over all D is $D(y) = \mathbb{E}[X | Y = y]$, and Proposition 2.16 says $D(y) = y + \sigma^2 \nabla \log p_Y(y)$. Hence

$$\nabla \log p_Y(y) = \frac{D(y) - y}{\sigma^2},$$

and conversely. Neither direction mentions p_X .

Fitting D requires only pairs $(X, X + \sigma\xi)$, which can be generated from samples of X by adding noise; neither p_X nor a normalizing constant appears. Module II.11 uses the score at every noise level σ to run the noising backwards.

Lectures 3 and 4

§1.2 What “converging to a Gaussian” means

1.3 C

The hypothesis already bounds how much mass X_n puts far out: test it against a **smooth cutoff**. Tightness is free, and after that only a mollification is left.

Let χ be smooth with $0 \leq \chi \leq 1$, $\chi = 1$ on $\{|x| \leq 1\}$ and $\chi = 0$ on $\{|x| \geq 2\}$, and put $\chi_M(x) = \chi(x/M)$, which is in the test class for each M . Since $\mathbf{1}\{|X| > M\} \rightarrow 0$ pointwise as $M \rightarrow \infty$ and is bounded by 1, dominated convergence gives $\mathbb{P}(|X| > M) \rightarrow 0$; choose M with $\mathbb{P}(|X| > M) < \varepsilon$. Then $\mathbb{E}\chi_M(X) \geq \mathbb{P}(|X| \leq M) > 1 - \varepsilon$, so

$$\limsup_n \mathbb{E}[1 - \chi_M(X_n)] = 1 - \mathbb{E}\chi_M(X) < \varepsilon.$$

Now let f be bounded continuous and set $g = f\chi_M$, continuous with support in $\{|x| \leq 2M\}$ and therefore uniformly continuous. For $g \sim \mathcal{N}(0, I_k)$ put $g_\delta(x) = \mathbb{E}[g(x + \delta g)] = \int g(y)\phi_\delta(x - y)dy$. Differentiating under the integral — the difference quotients of ϕ_δ are dominated by an integrable function of y — each derivative $\partial^\alpha g_\delta(x) = \int g(y)\partial^\alpha \phi_\delta(x - y)dy$ is bounded by $\|g\|_\infty \|\partial^\alpha \phi_\delta\|_{L^1}$, so g_δ lies in the test class; and with ω the modulus of continuity of g ,

$$|g_\delta(x) - g(x)| \leq \mathbb{E}|g(x + \delta g) - g(x)| \leq \mathbb{E}[\min(\omega(\delta|g|), 2\|g\|_\infty)] \xrightarrow{\delta \rightarrow 0} 0$$

uniformly in x . Finally

$$|\mathbb{E}f(X_n) - \mathbb{E}f(X)| \leq \|f\|_\infty \mathbb{E}[1 - \chi_M(X_n)] + \|f\|_\infty \mathbb{E}[1 - \chi_M(X)] + 2\|g - g_\delta\|_\infty + |\mathbb{E}g_\delta(X_n) - \mathbb{E}g_\delta(X)|.$$

Take $\limsup_n$: the last term vanishes by hypothesis and the first two are at most $2\|f\|_\infty \varepsilon$. Let $\delta \rightarrow 0$, then $\varepsilon \rightarrow 0$.

§1.6 Lindeberg and Feller

1.13 A Here $s_n^2 = n\sigma^2$ and every row entry has the law of X_1 , so

$$L_n(\varepsilon) = \frac{1}{n\sigma^2} \sum_{i=1}^n \mathbb{E}[X_i^2 \mathbf{1}\{|X_i| > \varepsilon\sigma\sqrt{n}\}] = \frac{1}{\sigma^2} \mathbb{E}[X_1^2 \mathbf{1}\{|X_1| > \varepsilon\sigma\sqrt{n}\}].$$

The integrand tends to 0 pointwise and is dominated by X_1^2 , which is integrable; dominated convergence gives $L_n(\varepsilon) \rightarrow 0$. So Theorem 1.9 applies with two moments only, where Theorem 1.7 assumed three. The third moment is not wasted: it is what buys the rate $n^{-1/2}$, which Theorem 1.9 does not supply.

1.14 C

Feller's condition is blind to the rare large value; Lindeberg's sees only that. The array is a Gaussian and a Poisson superimposed, and the limit records both.

$\text{Var}(X_{n,i}) = 1 \cdot (1 - \frac{1}{n}) + n \cdot \frac{1}{n} = 2 - \frac{1}{n}$, so $s_n^2 = 2n - 1$ and

$$\max_i \frac{\text{Var}(X_{n,i})}{s_n^2} = \frac{2 - \frac{1}{n}}{2n - 1} \rightarrow 0 :$$

Feller's condition holds. For $\varepsilon \leq 1/\sqrt{2}$ we have $\varepsilon^2 \leq \frac{1}{2} < \frac{n}{2n-1}$, i.e. $\varepsilon s_n < \sqrt{n}$, while $\varepsilon s_n > 1$ once n is large; so $\{|X_{n,i}| > \varepsilon s_n\} = \{|X_{n,i}| = \sqrt{n}\}$ and

$$L_n(\varepsilon) = \frac{1}{2n-1} \cdot n \cdot \left(n \cdot \frac{1}{n}\right) = \frac{n}{2n-1} \rightarrow \frac{1}{2}.$$

Lindeberg's condition fails. Feller's holds, so Remark 1.10(c) applies in contrapositive form: the array does *not* satisfy the central limit theorem. This is the array promised in Remark 1.10(b).

Directly: $\varphi_{n,i}(t) = (1 - \frac{1}{n}) \cos t + \frac{1}{n} \cos(t\sqrt{n})$, so with $s_n = \sqrt{2n-1}$,

$$\varphi_{T_n}(t) = \left[\left(1 - \frac{1}{n}\right) \cos \frac{t}{s_n} + \frac{1}{n} \cos \frac{t\sqrt{n}}{s_n} \right]^n = \left[1 - \frac{1}{n} \left(\frac{t^2}{4} + 1 - \cos \frac{t}{\sqrt{2}} \right) + o\left(\frac{1}{n}\right) \right]^n,$$

using $\cos(t/s_n) = 1 - \frac{t^2}{2(2n-1)} + O(n^{-2})$ and $t\sqrt{n}/s_n \rightarrow t/\sqrt{2}$. Hence

$$\varphi_{T_n}(t) \rightarrow \exp\left(-\frac{t^2}{4}\right) \cdot \exp\left(\cos \frac{t}{\sqrt{2}} - 1\right),$$

which is $\mathcal{N}(0, \frac{1}{2})$ convolved with a compound Poisson of rate 1 and jumps $\pm 1/\sqrt{2}$. Half the variance came from the ± 1 's and became Gaussian; the other half came from the rare $\pm\sqrt{n}$'s and did not. At $t = \sqrt{2}\pi$ the limit is $e^{-\pi^2/2-2}$ against $e^{-\pi^2/2}$ for the standard Gaussian.

§1.7 Records

1.21 A The array $X_{n,i} = B_{n,i} - p_{n,i}$ is independent and centered with $|X_{n,i}| \leq 1$ and $s_n = \sigma_n \rightarrow \infty$, so Lemma 1.15 gives Lindeberg's condition and Theorem 1.9 the conclusion. For $p_{n,i} = \lambda/n$ the total variance is $\sigma_n^2 = \lambda(1 - \lambda/n) \rightarrow \lambda$, which is bounded: the hypothesis $s_n \rightarrow \infty$ of Definition 1.8 fails, so nothing above applies, and the limit is Poisson(λ) — Exercise 1.33, or Theorem 2.10 with a rate.

1.22 B

Weak convergence survives affine maps that converge to the identity, provided no mass escapes to infinity. A uniform second moment is what rules that out.

The tool. Let f be bounded and uniformly continuous with modulus of continuity ω_f , and put $\delta_n = |a_n - 1|M + |b_n|$. On $\{|Z_n| \leq M\}$ the two arguments differ by at most δ_n , so the integrands differ by at most $\omega_f(\delta_n)$; off that event bound the difference by $2\|f\|_\infty$ and use $\mathbb{P}(|Z_n| > M) \leq \mathbb{E}Z_n^2/M^2 \leq C/M^2$. Hence

$$|\mathbb{E}f(a_n Z_n + b_n) - \mathbb{E}f(Z_n)| \leq \omega_f(\delta_n) + \frac{2\|f\|_\infty C}{M^2}.$$

Let $n \rightarrow \infty$ with M fixed, then $M \rightarrow \infty$: the left side tends to 0, and $\mathbb{E}f(Z_n) \rightarrow \mathbb{E}f(Z)$ by hypothesis.

The application. With $Z_n = (R_n - H_n)/s_n$, which has $\mathbb{E}Z_n^2 = 1$,

$$\frac{R_n - \log n}{\sqrt{\log n}} = \frac{s_n}{\sqrt{\log n}} Z_n + \frac{H_n - \log n}{\sqrt{\log n}},$$

and $s_n^2/\log n = (H_n - \sum_{i \leq n} i^{-2})/\log n \rightarrow 1$ while $H_n - \log n \rightarrow \gamma$, so the numerator of the second term is bounded. Theorem 1.17 and the tool give the claim.

Why Lemma 1.2 is not enough. Lemma 1.2 upgrades convergence against a *fixed* family of test functions. Here the quantity to control is $\mathbb{E}f(a_n Z_n + b_n)$, which is $\mathbb{E}f_n(Z_n)$ for a test function f_n that moves with n ; convergence for each fixed f says nothing about it directly. The uniform second moment is what closes the gap.

1.23 B Each sequence of choices $(c_1, \dots, c_n)$ with $c_i \in \{1, \dots, i\}$ has probability $\prod_{i \leq n} \frac{1}{i} = \frac{1}{n!}$, and there are $n!$ of them. The map to permutations is injective — from the finished permutation, delete n from its cycle and read off what it did, then recurse — hence a bijection onto S_n , so the construction produces the uniform law. Element i opens a new cycle exactly when its choice is the distinguished one, which happens with probability $1/i$, and the choices are independent by construction. So K_n has the law of $\sum_{i \leq n} \text{Bernoulli}(1/i)$ with independent summands, which is the array of Theorem 1.17; Exercise 1.22 supplies the centring by $\log n$.

§1.8* A wide network at initialization

1.25 B $\mathbb{E}V_j = \mathbb{E}[a_j] \mathbb{E}[\sigma(\langle w_j, x_i \rangle)] = 0$: independence of a_j from w_j is what lets the expectation factor, and $\mathbb{E}a_j = 0$ kills it. Likewise $\text{Cov}(V_j)_{il} = \mathbb{E}[a_j^2] \mathbb{E}[\sigma(\langle w_j, x_i \rangle)\sigma(\langle w_j, x_l \rangle)] = K(x_i, x_l)$, the same factoring with $\mathbb{E}a_j^2 = \sigma_a^2$. The V_j are i.i.d. because the pairs (a_j, w_j) are, and $(f_m(x_1), \dots, f_m(x_k)) = m^{-1/2} \sum_j V_j$ by definition.

The multidimensional central limit theorem then applies with only a second moment. The second route is more elementary but assumes more. For fixed $\xi \in \mathbb{R}^k$, Theorem 1.7 applied to the scalars $\langle \xi, V_j \rangle$ gives $\langle \xi, m^{-1/2} \sum_j V_j \rangle \Rightarrow \mathcal{N}(0, \xi^\top K \xi)$, hence $\varphi_{m^{-1/2} \sum_j V_j}(\xi) = \mathbb{E}[e^{-i\langle \xi, \cdot \rangle}] \rightarrow e^{-\frac{1}{2}\xi^\top K \xi}$ for every ξ , and Lemma 1.29 finishes. But Theorem 1.7 is stated with $\mathbb{E}|X_1|^3 < \infty$, so this route needs a third moment.

1.26 A $\sum_{i,l} v_i v_l K(x_i, x_l) = \sigma_a^2 \mathbb{E}[(\sum_i v_i \sigma(\langle w, x_i \rangle))^2] \geq 0$. K is a covariance because it *is* one — of V_j — so no condition on σ beyond square-integrability could have been needed.

1.27 C

$(\langle w, x \rangle, \langle w, x' \rangle)$ is a centered Gaussian pair, and by Theorem 1.5 of Lectures 1–2 its law depends on x, x' only through the 2×2 Gram matrix. **The d -dimensional integral is two-dimensional.**

σ is positively homogeneous, so K scales as $\|x\|\|x'\|$ and we may take $\|x\| = \|x'\| = 1$. The pair $(\langle w, x \rangle, \langle w, x' \rangle)$ is centered Gaussian with covariance $\begin{pmatrix} 1 & \cos \theta \\ \cos \theta & 1 \end{pmatrix}$, hence equal in law to $(g_1, \cos \theta g_1 + \sin \theta g_2)$ for independent standard g_1, g_2 .

Write $(g_1, g_2) = R(\cos \alpha, \sin \alpha)$ with $R \geq 0$ and α uniform on $[0, 2\pi)$ independent, $\mathbb{E}R^2 = 2$. Then the pair is $(R \cos \alpha, R \cos(\alpha - \theta))$ and

$$\mathbb{E}[\sigma \sigma'] = \mathbb{E}[R^2] \cdot \frac{1}{2\pi} \int_0^{2\pi} (\cos \alpha)_+ (\cos(\alpha - \theta))_+ d\alpha = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos \alpha \cos(\alpha - \theta) d\alpha,$$

the integrand being positive exactly on that interval, of length $\pi - \theta$. Now $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$ gives

$$\int_{\theta-\pi/2}^{\pi/2} \cos \alpha \cos(\alpha - \theta) d\alpha = \frac{1}{2}(\pi - \theta) \cos \theta + \frac{1}{4} \left[\sin(2\alpha - \theta) \right]_{\theta-\pi/2}^{\pi/2} = \frac{1}{2}[(\pi - \theta) \cos \theta + \sin \theta],$$

since the bracket evaluates to $\sin \theta - (-\sin \theta) = 2 \sin \theta$. Hence the value is $\frac{1}{2\pi}(\sin \theta + (\pi - \theta) \cos \theta)$.

Checks. At $\theta = 0$: $\frac{1}{2\pi} \cdot \pi = \frac{1}{2} = \mathbb{E}[\max(0, g)^2]$. At $\theta = \pi/2$ the two projections are independent, so the answer should be $\mathbb{E}[\max(0, g)]^2 = (2\pi)^{-1}$, and it is.

§1.9* The proof we did not give

1.30 A The j th difference quotient of $t \mapsto e^{-itX}$ is dominated by $|X|^j$, integrable for $j \leq k$, so we may differentiate $\varphi_X(t) = \mathbb{E}[e^{-itX}]$ under the expectation k times, bringing down $(-iX)^k$. For $\varphi = e^{-t^2/2}$: $\varphi' = -t\varphi$, $\varphi'' = (t^2 - 1)\varphi$, so $\varphi''(0) = -1 = (-i)^2 \mathbb{E}g^2$. The series $e^{-t^2/2} = 1 - \frac{t^2}{2} + \frac{t^4}{8} - \dots$ has t^4 coefficient $\frac{1}{8}$, so $\varphi^{(4)}(0) = 4!/8 = 3 = (-i)^4 \mathbb{E}g^4$.

1.32 B With $a = 1 + z_n/n$ and $b = e^{z_n/n}$, both are $1 + O(1/n)$ so $|a|, |b| \leq M$ for some fixed M , and $|a - b| = O(1/n^2)$; hence $|a^n - b^n| \leq nM^{n-1}|a - b| \rightarrow 0$ while $b^n = e^{z_n} \rightarrow e^z$. Applying this with $z_n = n(\varphi_{X_1}(t/\sqrt{n}) - 1) \rightarrow -t^2/2$ by Lemma 1.31 gives $\varphi_{S_n}(t) \rightarrow e^{-t^2/2}$, and Lemma 1.29 finishes. **Independence** is what gives $\varphi_{S_n} = \prod_i \varphi_{X_i}(t/\sqrt{n})$; **identical distribution** only lets us write one factor n times.

1.33 A $\varphi_B(t) = (1 - p) + pe^{-it} = 1 + p(e^{-it} - 1)$. So $\varphi_{N_n} = [1 + \frac{\lambda}{n}(e^{-it} - 1)]^n \rightarrow \exp(\lambda(e^{-it} - 1))$ by Exercise 1.32, and $\sum_k e^{-\lambda \frac{\lambda^k}{k!}} e^{-itk} = e^{-\lambda} e^{\lambda e^{-it}}$ is the same thing.

1.34 B $X_{n,i}$ takes $1 - \lambda/n$ with probability λ/n and $-\lambda/n$ otherwise, so $s_n^2 = n \cdot \frac{\lambda}{n}(1 - \frac{\lambda}{n}) = \lambda(1 - \frac{\lambda}{n})$. For $\varepsilon < 1/\sqrt{\lambda}$ the threshold $\varepsilon s_n \rightarrow \varepsilon\sqrt{\lambda} < 1$ eventually falls below the large value, so $L_n(\varepsilon) \rightarrow n \cdot \frac{\lambda}{n} \cdot 1/\lambda = 1$.

At $\lambda = 3$, $\varepsilon = \frac{1}{2}$: $n = 10$ gives $s_n = 1.449$ and $\varepsilon s_n = 0.7246$, while the large value is $1 - \frac{3}{10} = 0.7 < 0.7246$ — nothing is caught, so $L_{10} = 0$ **exactly**. Then 0.970 and 1.000.

1.35 C

A Gaussian characteristic function $e^{-\sigma^2 t^2/2}$ **never vanishes**, and weak convergence forces the transforms to converge. So one zero rules out every Gaussian limit, with no need to know that a limit exists.

$s_n^2 = n \cdot 1 + n = 2n$. The large summand has $|X_{n,n+1}| = \sqrt{n} > \varepsilon\sqrt{2n} = \varepsilon s_n$ exactly when $\varepsilon < 1/\sqrt{2}$, and it alone contributes $n/2n = \frac{1}{2}$ to L_n . The first n summands give $s_n^{-1} \sum_{i \leq n} X_{n,i} \Rightarrow \mathcal{N}(0, \frac{1}{2})$ with transform $e^{-t^2/4}$; the last gives $\pm 1/\sqrt{2}$ with transform $\cos(t/\sqrt{2})$; they are independent, so $\varphi_n(t) \rightarrow e^{-t^2/4} \cos(t/\sqrt{2})$, which vanishes at $t = \pi/\sqrt{2}$. If $s_n^{-1} \sum_i X_{n,i} \Rightarrow \mathcal{N}(0, \sigma^2)$ then, $e^{-it \cdot}$ being bounded and continuous, $\varphi_n(t) \rightarrow e^{-\sigma^2 t^2/2}$ for every t ; at $t = \pi/\sqrt{2}$ that is impossible.

1.36 C

S_n has an **atom**. A distribution function with a jump of size a is at least $a/2$ from any continuous one.

If $F(z_0) - F(z_0^-) = a$ and Φ is continuous then $|F(z_0) - \Phi(z_0)| + |F(z_0^-) - \Phi(z_0^-)| \geq a$, so the sup is $\geq a/2$. For $X_i = \pm 1$ and n even, $\mathbb{P}(S_n = 0) = \binom{n}{n/2} 2^{-n} \sim \sqrt{2/\pi n}$, giving $\sup_z |\mathbb{P}(S_n \leq z) - \Phi(z)| \geq c n^{-1/2}$. Lattice summands were slowest in §1.6 for exactly this reason: their atoms cannot be smoothed, and they are of the same order as the whole error budget.

1.37 C

Compare $1 - \varphi_Y$ against $1 - \frac{\sin x}{x}$, which is non-negative everywhere and at least $\frac{1}{2}$ once $|x| \geq 2$.

Since $\varphi_Y(-t) = \overline{\varphi_Y(t)}$ the integral is real, and by Fubini — the integrand is bounded by 1 on a bounded interval — $\int_{-u}^u \varphi_Y(t) dt = \mathbb{E} \int_{-u}^u e^{-itY} dt = \mathbb{E} \left[\frac{2 \sin(uY)}{Y} \right]$, so

$$\frac{1}{u} \int_{-u}^u (1 - \varphi_Y(t)) dt = 2 \mathbb{E} \left[1 - \frac{\sin(uY)}{uY} \right].$$

The bracket is non-negative because $|\sin x| \leq |x|$, and at least $\frac{1}{2}$ when $|uY| \geq 2$ because $|\sin x/x| \leq 1/|x|$. Hence the right side is at least $2 \cdot \frac{1}{2} \mathbb{P}(|uY| \geq 2) = \mathbb{P}(|Y| \geq 2/u)$.

For tightness, $\psi(0) = \lim_n \varphi_{Y_n}(0) = 1$, so continuity of ψ at 0 gives, for any $\varepsilon > 0$, a $u > 0$ with $\frac{1}{u} \int_{-u}^u (1 - \psi) < \varepsilon/2$. The integrands are bounded by 2, so dominated convergence gives $\frac{1}{u} \int_{-u}^u (1 - \varphi_{Y_n}) \rightarrow \frac{1}{u} \int_{-u}^u (1 - \psi)$ and $\mathbb{P}(|Y_n| \geq 2/u) < \varepsilon$ for all $n \geq N$. Increasing K past $2/u$ covers the finitely many remaining n .

§2.1 The characterization

2.3 B

Both sides are integrals of $\hat{f}$ against the Gaussian transform. On that side multiplication by X_j is a derivative in ξ , and the Gaussian transform is its own derivative up to the factor $-(\Sigma\xi)_j$; the two integrands become equal. Nothing is inverted, so Σ may be singular.

(a) Put $\psi(\xi) = \mathbb{E}[e^{i\langle X, \xi \rangle}]$, which by Definition 1.3 of Lectures 1–2 equals $e^{-\frac{1}{2}\xi^\top \Sigma \xi}$ for every $\Sigma \succeq 0$. By Fourier inversion $f(x) = (2\pi)^{-d} \int \hat{f}(\xi) e^{i\langle x, \xi \rangle} d\xi$, and since $\mathbb{E} \int |\hat{f}(\xi) X_j| d\xi = \mathbb{E}|X_j| \|\hat{f}\|_{L^1} < \infty$, Fubini gives

$$\mathbb{E}[X_j f(X)] = \frac{1}{(2\pi)^d} \int \hat{f}(\xi) \mathbb{E}[X_j e^{i\langle X, \xi \rangle}] d\xi.$$

Differentiating under the expectation, $\partial_{\xi_j} \psi(\xi) = \mathbb{E}[i X_j e^{i\langle X, \xi \rangle}]$, while differentiating the formula for ψ gives $\partial_{\xi_j} \psi(\xi) = -(\Sigma\xi)_j \psi(\xi)$. Hence $\mathbb{E}[X_j e^{i\langle X, \xi \rangle}] = i(\Sigma\xi)_j \psi(\xi)$.

Since $\xi_k \hat{f} \in L^1$, the function $\partial_k f$ is recovered by inversion from $\widehat{\partial_k f} = i \xi_k \hat{f}$, and the same Fubini gives

$$\mathbb{E}[\partial_k f(X)] = \frac{1}{(2\pi)^d} \int i \xi_k \hat{f}(\xi) \psi(\xi) d\xi.$$

Multiply by Σ_{jk} and sum over k : the integrand becomes $i(\Sigma\xi)_j \hat{f}(\xi) \psi(\xi)$, which is the integrand of the first display. Only the formula $\psi(\xi) = e^{-\frac{1}{2}\xi^\top \Sigma \xi}$ was used; no square root or inverse of Σ appears, and the degenerate case needs no separate treatment.

(b) Fix v and take $F(x) = v f(\langle v, x \rangle)$ with $f \in C^1$ and f, f' bounded. Then $\langle W, F(W) \rangle = Y f(Y)$ for $Y = \langle v, W \rangle$, and $DF(x) = f'(\langle v, x \rangle) v v^\top$, so $\text{tr}(\Sigma DF) = \sigma^2 f'(Y)$ with $\sigma^2 = v^\top \Sigma v$. The hypothesis reads $\mathbb{E}[Y f(Y)] = \sigma^2 \mathbb{E}[f'(Y)]$. For $\sigma > 0$ this is Theorem 2.1 for Y/σ , so $Y \sim \mathcal{N}(0, \sigma^2)$; for $\sigma = 0$ take $f(y) = y/(1 + y^2)$, which is admissible, to get $\mathbb{E}[Y^2/(1 + Y^2)] = 0$ and hence $Y = 0$. Every linear functional of W is Gaussian, so $W \sim \mathcal{N}(0, \Sigma)$ by Corollary 1.7 of Lectures 1–2.

(c) Exercise 2.3(a) with $\{(k, l)\}$ as the index set: a matrix is a vector whose index is a pair, with covariance $\Sigma_{(ij), (kl)} = \mathbb{E}[M_{ij} M_{kl}]$. Nothing in that argument used the shape of the index set.

§2.5* Where independence was spent

2.13 A $\mathbb{E}F(W, W') = \mathbb{E}F(W', W)$ by exchangeability, and $= -\mathbb{E}F(W, W')$ by antisymmetry; so it is zero.

2.14 B $F(x, y) = (y - x)(f(x) + f(y))$ is antisymmetric, so $\mathbb{E}[(W' - W)(f(W) + f(W'))] = 0$. Writing $f(W') = f(W) + (W' - W)f'(W) + R$ with $|R| \leq \frac{\|f''\|_\infty}{2} (W' - W)^2$,

$$0 = 2\mathbb{E}[(W' - W)f(W)] + \mathbb{E}[(W' - W)^2 f'(W)] + \mathbb{E}[(W' - W)R].$$

The first term is $-2a\mathbb{E}[Wf(W)]$ by the Stein-pair property and conditioning on W . Rearranging gives the display, with the remainder bounded by $\frac{\|f''\|_\infty}{4a} \mathbb{E}|W' - W|^3$.

2.15 B Swapping $X_I \leftrightarrow X'_I$ is measure preserving, so (W, W') is exchangeable. Since $\mathbb{E}[X'_I | X_1, \dots, X_n] = 0$ and $\mathbb{E}[X_I | X_1, \dots, X_n] = \frac{1}{n} \sum_i X_i$, $\mathbb{E}[W' - W | X_1, \dots, X_n] = -\frac{1}{\sqrt{n}} \cdot \frac{1}{n} \sum_i X_i = -W/n$, and the tower property gives $a = 1/n$. Also $(W' - W)^2 = (X_I - X'_I)^2/n$, whose conditional expectation is $\frac{1}{n^2} \sum_i (X_i^2 + 1)$. Dividing by $2a = 2/n$ gives $\frac{1}{2}(1 + \frac{1}{n} \sum_i X_i^2)$, whose distance from 1 is $\frac{1}{2}|1 - \frac{1}{n} \sum_i X_i^2|$, at most $\frac{1}{2}\sqrt{\text{Var}(X_1^2)/n}$ in L^1 . The remainder is $\frac{n\|f''\|_\infty}{4} n^{-3/2} \mathbb{E}|X_1 - X'_1|^3$. Both are $O(n^{-1/2})$.

Where independence entered: in the two conditional expectations, and nowhere else. The product formula $\mathbb{E}[X_i f(W^{(i)})] = \mathbb{E}[X_i] \mathbb{E}[f(W^{(i)})]$ never appeared.

2.16 A $\mathbb{E}W = n \cdot \frac{1}{n} = 1$; $\mathbb{E}[I_i I_j] = \frac{1}{n(n-1)}$ for $i \neq j$ gives $\mathbb{E}W^2 = 1 + 1 = 2$, so $\text{Var } W = 1$. And $\mathbb{P}(I_1 = I_2 = 1) = \frac{1}{n(n-1)} \neq \frac{1}{n^2}$. What fails is **independence of the I_i** , which Theorem 2.10 requires; the theorem therefore says nothing, and a true conclusion reached from unsatisfied hypotheses is still not proved.

2.17 C

Force i to be a fixed point by **composing with the transposition that sends i to itself**: $\sigma = \pi \circ (i \ \pi^{-1}(i))$.

Then $\sigma(i) = \pi(\pi^{-1}(i)) = i$, and the map is a bijection from $\{\pi : \pi(i) \neq i\}$ onto itself composed with a relabelling, so σ is uniform among permutations fixing i . Writing $j = \pi^{-1}(i)$: i becomes fixed (gain 1), and j becomes fixed exactly when $\pi(i) = j$, i.e. when π has a 2-cycle at i (gain 2). Hence $V_i = W(\sigma) - 1 \in \{W, W + 1\}$, and

$$\mathbb{E}|W - V_i| \leq \mathbb{P}(\pi(i) = i) + \mathbb{P}(\text{2-cycle at } i) = \frac{1}{n} + \frac{1}{n}.$$

Chen–Stein with $p_i = 1/n$, $\lambda = 1$ gives $d_{TV}(W, \text{Poisson}(1)) \leq n \cdot \frac{1}{n} \cdot \frac{2}{n} = \frac{2}{n}$.

At $n = 10$ the bound is 0.2 and the truth is 2.3×10^{-8} : wrong by about seven orders of magnitude. What it has that inclusion–exclusion does not is **robustness** — it never used that π was a permutation beyond the coupling, and it is the only one of the three routes in this course that produces a bound at all.

Lectures 5–8

§1.3 The Gaussian Wigner matrix

1.9 A $M_{ij} = \frac{1}{\sqrt{2N}}(A_{ij} + A_{ji})$ with the A_{ab} i.i.d. standard, so $\mathbb{E}[A_{ab}A_{cd}] = \delta_{ac}\delta_{bd}$ and

$$\mathbb{E}[M_{ij}M_{kl}] = \frac{1}{2N}\mathbb{E}[(A_{ij} + A_{ji})(A_{kl} + A_{lk})] = \frac{1}{2N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} + \delta_{il}\delta_{jk} + \delta_{ik}\delta_{jl}),$$

which is $\frac{1}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$. At $i = j = k = l$ both delta terms fire and the formula returns $\frac{2}{N}$, so the diagonal variance is not a convention imposed on the model but a consequence of symmetrizing A : the diagonal entry $M_{ii} = \sqrt{2/N} A_{ii}$ collects the same Gaussian twice.

§1.5* Cumulants

1.14 B Write $K(t) = \log \mathbb{E}[e^{\sum_i t_i Y_i}]$.

(a) Symmetry is the equality of mixed partials. For linearity let $\tilde{K}(s, s', t_2, \dots, t_m) = \log \mathbb{E} \exp(sY + s'Z + \sum_{i \geq 2} t_i Y_i)$. The log-transform of $(aY + bZ, Y_2, \dots, Y_m)$ is $K(t_1, \dots, t_m) = \tilde{K}(at_1, bt_1, t_2, \dots, t_m)$, so by the chain rule, applied once since t_1 is differentiated once,

$$\partial_{t_1} K = a \partial_s \tilde{K} + b \partial_{s'} \tilde{K}.$$

Now apply $\partial_{t_2} \cdots \partial_{t_m}$ and set every variable to 0: the two terms are $a\kappa(Y, Y_2, \dots)$ and $b\kappa(Z, Y_2, \dots)$. Adding a constant c to Y_1 adds ct_1 to K , which any ∂_{t_j} with $j \geq 2$ kills. Applying linearity in each of the m slots gives $\kappa(cY, \dots, cY) = c^m \kappa(Y, \dots, Y)$.

(b) If the arguments split into independent groups $B_1 \sqcup B_2$, then $\mathbb{E}[e^{\sum t_i Y_i}]$ factors and $K = F_1(t_{B_1}) + F_2(t_{B_2})$. The derivative $\partial_{t_1} \cdots \partial_{t_m}$ contains a derivative in each group, so it kills both summands. For additivity expand $\kappa(Y + Z, \dots, Y + Z)$ by (a) into 2^m terms $\kappa(W_1, \dots, W_m)$ with each $W_j \in \{Y, Z\}$; every term containing both Y and Z vanishes, leaving $\kappa(Y, \dots, Y) + \kappa(Z, \dots, Z)$.

(c) Definition 1.3 of Lectures 1–2 at $\xi = it$ gives $K(t) = \frac{1}{2}t^\top \Sigma t$, a quadratic form: its first derivatives vanish at 0, its second are Σ_{ab} , and all higher ones are identically 0.

1.15 B

Multilinearity expands $\kappa(S_n, \dots, S_n)$ into n^m joint cumulants of the X_i , and independence **kills every one that is not on the diagonal**.

By Exercise 1.14(a) we may drop the centering and pull $n^{-1/2}$ out of each of the m slots:

$$\kappa(S_n, \dots, S_n) = n^{-m/2} \sum_{i_1, \dots, i_m=1}^n \kappa(X_{i_1}, \dots, X_{i_m}).$$

If the indices are not all equal, the arguments split into those equal to X_{i_1} and the rest, a non-empty group independent of the first, so that term vanishes by Exercise 1.14(b). The n diagonal terms each equal $\kappa(X_1, \dots, X_1)$, so

$$\kappa(S_n, \dots, S_n) = n \cdot n^{-m/2} \kappa(X_1, \dots, X_1) = n^{1-m/2} \kappa(X_1, \dots, X_1).$$

At $m = 2$ the exponent is 0 and $\kappa(S_n, S_n) = \kappa(X_1, X_1) = \sigma^2$ for every n . For $m \geq 3$ it is negative, so every higher cumulant tends to 0: in the limit only the second cumulant survives, which is exactly what a Gaussian looks like.

§2.3 Two moments by hand

2.6 A $\text{tr } M^{2k+1}$ is a sum of products of an odd number of centered jointly Gaussian variables, and Theorem 1.2 pairs factors two at a time, so every term has mean 0. (Alternatively, $-M$ has the same law as M , so $\text{tr } M^{2k+1}$ is a symmetric random variable.) Next, $\text{tr } M = \sum_i M_{ii}$ is a sum of N independent centered variables of variance $\frac{2}{N}$, so $\mathbb{E}[(\text{tr } M)^2] = 2$.

§2.4–§2.7 Paths, every moment, and the fluctuations

2.20 A Of the $5!! = 15$ pair partitions of $\{1, \dots, 6\}$, ten contain a crossing pair (for example $\{13\}\{24\}\{56\}$, crossing at $1 < 2 < 3 < 4$), and five are non-crossing. For each of those five, a tree path with that pairing, starting from a vertex r :

$\{12\}\{34\}\{56\}$	$r \rightarrow a \rightarrow r \rightarrow b \rightarrow r \rightarrow c \rightarrow r$	three edges from r
$\{12\}\{36\}\{45\}$	$r \rightarrow a \rightarrow r \rightarrow b \rightarrow c \rightarrow b \rightarrow r$	one edge, then a path of two
$\{14\}\{23\}\{56\}$	$r \rightarrow a \rightarrow b \rightarrow a \rightarrow r \rightarrow c \rightarrow r$	a path of two, then one edge
$\{16\}\{23\}\{45\}$	$r \rightarrow a \rightarrow b \rightarrow a \rightarrow c \rightarrow a \rightarrow r$	one edge, then two excursions from its end
$\{16\}\{25\}\{34\}$	$r \rightarrow a \rightarrow b \rightarrow c \rightarrow b \rightarrow a \rightarrow r$	a path of three, out and back

Each visits r, a, b, c : four vertices, $k+1$ for $k=3$.

2.21 B ρ_{sc} is even, so its odd moments vanish. Substituting $x = 2 \sin \theta$ gives $\sqrt{4-x^2} = 2 \cos \theta$ and $dx = 2 \cos \theta d\theta$ on $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, so

$$\int_{-2}^2 x^{2k} \rho_{sc}(x) dx = \frac{4^{k+1}}{2\pi} \int_{-\pi/2}^{\pi/2} \sin^{2k} \theta \cos^2 \theta d\theta.$$

At $k=0$ the integral is $\int \cos^2 = \frac{\pi}{2}$, giving $\frac{4}{2\pi} \cdot \frac{\pi}{2} = 1$: ρ_{sc} is a probability density. At $k=1$, $\int_{-\pi/2}^{\pi/2} \sin^2 \theta \cos^2 \theta d\theta = \frac{1}{8} \int_{-\pi/2}^{\pi/2} \sin^2 2\theta d\theta = \frac{\pi}{8}$, giving $\frac{16}{2\pi} \cdot \frac{\pi}{8} = 1 = C_1$. At $k=2$, $\int_{-\pi/2}^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta = \frac{\pi}{16}$, giving $\frac{64}{2\pi} \cdot \frac{\pi}{16} = 2 = C_2$.

§3 The Stieltjes transform

3.11 A (a) $M=0$: every $\lambda_i = 0$, so $m_N(z) = -\frac{1}{z}$. (b) Half at ± 1 :

$$m_N(z) = \frac{1}{2} \left(\frac{1}{-1-z} + \frac{1}{1-z} \right) = \frac{-z}{z^2-1}.$$

In both cases $|\lambda_i - z| \geq \Im z$ for real λ_i , so $|m_N(z)| \leq \frac{1}{\Im z}$, which is (7). And $\frac{1}{\lambda-z} = -\frac{1}{z} \left(1 - \frac{\lambda}{z}\right)^{-1} = -\frac{1}{z} - \frac{\lambda}{z^2} - \dots$, averaged over i , gives $m_N(z) = -\frac{1}{z} - \frac{\text{tr } M/N}{z^2} + O(|z|^{-3})$; here $\text{tr } M = 0$ in both cases, so the second term drops out.

3.12 A At $\varepsilon = 1$ the curve is a broad bump — the semicircle smoothed by a Cauchy kernel of width 1, too smooth to show the edges at ± 2 . At $\varepsilon = 0.1$ it tracks ρ_{sc} closely, edges included: the kernel still averages over $\varepsilon N \approx 200$ eigenvalues. At $\varepsilon = 0.001$ the individual eigenvalues resolve into spikes of height $\sim \frac{1}{\pi N \varepsilon}$. The crossover is at $\varepsilon \sim 1/N$, the mean spacing in the bulk: above it the resolvent sees the law, below it the sample.

3.13 B With $z = \lambda + i\varepsilon$,

$$\frac{1}{\lambda_i - z} = \frac{1}{(\lambda_i - \lambda) - i\varepsilon} = \frac{(\lambda_i - \lambda) + i\varepsilon}{(\lambda_i - \lambda)^2 + \varepsilon^2},$$

whose imaginary part is $\frac{\varepsilon}{(\lambda_i - \lambda)^2 + \varepsilon^2} > 0$. Averaging over i gives the display, and positivity of every term gives $\Im m_N > 0$ on the upper half plane. Multiplying by ε ,

$$\varepsilon \Im m_N(\lambda + i\varepsilon) = \frac{1}{N} \sum_{i=1}^N \frac{\varepsilon^2}{(\lambda_i - \lambda)^2 + \varepsilon^2},$$

a finite sum in which each term equals 1 when $\lambda_i = \lambda$ and tends to 0 otherwise. So the limit is $\frac{1}{N}$ times the number of i with $\lambda_i = \lambda$; no convergence theorem is needed, the sum has N terms.

For the picture: the i th term of $\Im m_N$ is a bump centered at λ_i of height $\frac{1}{N\varepsilon}$ and width ε , so $\frac{1}{\pi} \Im m_N(\lambda + i\varepsilon)$ is roughly $\frac{1}{N\varepsilon}$ times the number of eigenvalues within ε of λ . In the bulk that number is about $N\rho_{sc}(\lambda)\varepsilon$, which is a stable average only when $N\varepsilon \gg 1$. For $\varepsilon \ll 1/N$ the nearest eigenvalue dominates and the graph is a comb of spikes of height $\frac{1}{\pi N\varepsilon}$. That is the crossover of Exercise 3.12.

§4 The semicircle law by the Stieltjes transform

4.22 A $A^{-1} = \frac{1}{\alpha d - b^2} \begin{pmatrix} d & -b \\ -b & \alpha \end{pmatrix}$. The corner is $\frac{d}{\alpha d - b^2} = \frac{1}{\alpha - b^2/d}$. The trace is $\frac{\alpha + d}{\alpha d - b^2}$, and the right side of the trace identity is

$$\frac{1}{d} + \frac{d}{\alpha d - b^2} \left(1 + \frac{b^2}{d^2}\right) = \frac{(\alpha d - b^2) + (d^2 + b^2)}{d(\alpha d - b^2)} = \frac{\alpha + d}{\alpha d - b^2}.$$

For $A = M - zI$ with M symmetric 2×2 , $g = (A^{-1})_{11} = (M_{11} - z - \frac{M_{12}^2}{M_{22} - z})^{-1}$ and $\text{tr } A^{-1} - \frac{1}{M_{22} - z} = g(1 + \frac{M_{12}^2}{(M_{22} - z)^2})$. With $t = \Im \frac{M_{12}^2}{M_{22} - z} = \frac{M_{12}^2 \Im z}{|M_{22} - z|^2} \geq 0$, the denominator of g has imaginary part $-\Im z - t$, so $|g| \leq \frac{1}{\Im z + t}$, while $|1 + \frac{M_{12}^2}{(M_{22} - z)^2}| \leq 1 + \frac{M_{12}^2}{|M_{22} - z|^2} = 1 + \frac{t}{\Im z}$. The product is at most $\frac{1+t/\Im z}{\Im z + t} = \frac{1}{\Im z}$, which is (14) for $N = 2$.

4.23 A If λ_i are the eigenvalues of M then $\sigma \lambda_i$ are those of σM , so the transform of σM is $\frac{1}{N} \sum_i \frac{1}{\sigma \lambda_i - z} = \frac{1}{\sigma} m_N(z/\sigma)$. Since $\Im(z/\sigma) = \Im z/\sigma > 0$, Theorem 4.1(b) applies at the point z/σ and gives $\mathbb{E}[\frac{1}{\sigma} m_N(z/\sigma)] \rightarrow \frac{1}{\sigma} m(z/\sigma) =: m_\sigma(z)$, uniformly on bounded sets of $\Re z$. Put $w = z/\sigma$ and $m(w) = \sigma m_\sigma(z)$ into $m(w)^2 + wm(w) + 1 = 0$: $\sigma^2 m_\sigma^2 + z m_\sigma + 1 = 0$. And $|m_\sigma| = |m|/\sigma < 1/\sigma$. Solving, $m_\sigma(z) = \frac{-z + \sqrt{z^2 - 4\sigma^2}}{2\sigma^2}$ with the root of positive imaginary part.

For the density, on the real axis the coefficients are real, so if $\Im m_\sigma \neq 0$ the two roots are conjugate and their product $1/\sigma^2$ gives $|\sigma m_\sigma| = 1$. Write $\sigma m_\sigma = e^{i\theta}$; dividing $\sigma^2 m_\sigma^2 + x m_\sigma + 1 = 0$ by $\sigma e^{i\theta}$ gives $\sigma e^{i\theta} + x + \sigma e^{-i\theta} = 0$, so $x = -2\sigma \cos \theta$ and $\Im m_\sigma = \frac{\sin \theta}{\sigma} = \frac{1}{\sigma} \sqrt{1 - \frac{x^2}{4\sigma^2}} = \frac{1}{2\sigma^2} \sqrt{4\sigma^2 - x^2}$. Dividing by π gives the density, supported on $[-2\sigma, 2\sigma]$. Proposition 3.7 then turns this into the proportion of eigenvalues of σM in every interval, exactly as in Theorem 4.19.

4.24 B The independent variables are M_{ij} , $i \leq j$. Since $f = \sum_{i,j} M_{ij}^2 = \sum_i M_{ii}^2 + 2 \sum_{i < j} M_{ij}^2$, we get $\partial f / \partial M_{ij} = 4M_{ij}$ for $i < j$ and $\partial f / \partial M_{ii} = 2M_{ii}$. The Poincaré sum is

$$\sum_{i < j} \frac{1}{N} \mathbb{E}(4M_{ij})^2 + \sum_i \frac{2}{N} \mathbb{E}(2M_{ii})^2 = \frac{16}{N} \binom{N}{2} \frac{1}{N} + \frac{8}{N} \cdot N \cdot \frac{2}{N} = 8 \left(1 - \frac{1}{N}\right) + \frac{16}{N} = 8 + \frac{8}{N},$$

and Lemma 4.2 multiplies it by $\frac{\pi^2}{8}$. The bound is $O(1)$ although $\text{tr } M^2$ is a sum of N^2 terms; the exact variance $4 + \frac{4}{N}$ is smaller by the factor $\frac{\pi^2}{2}$ and the crude constant. What Poincaré sees is that each of the N^2 partial derivatives is of size $N^{-1/2}$ and their squares sum to $O(1)$ — the same mechanism as in Proposition 4.3, with M in place of R^2 .

Lectures 9–11

§1 Gaussian processes

1.10 A (a) $(Mu)_i = \sum_k M_{ik} u_k$, so $\mathbb{E}[(Mu)_i (Mv)_j] = \sum_{k,l} u_k v_l \mathbb{E}[M_{ik} M_{jl}] = \sum_{k,l} u_k v_l \delta_{ij} \delta_{kl} = \delta_{ij} \langle u, v \rangle$. The pair (Mu, Mv) is a linear image of the Gaussian vector of entries of M , hence jointly Gaussian (Theorem 1.5 of Lectures 1–2), and $\mathbb{E} \langle Mu, Mv \rangle = \sum_i \mathbb{E}[(Mu)_i (Mv)_i] = N \langle u, v \rangle$.

(b) Cauchy–Schwarz: $\langle g, u \rangle \leq \|g\| \|u\| = \|g\|$ for unit u , with equality at $u = g/\|g\|$. $\mathbb{E}\|g\|^2 = \sum_i \mathbb{E}g_i^2 = d$, and Jensen gives $\mathbb{E}\|g\| \leq \sqrt{\mathbb{E}\|g\|^2} = \sqrt{d}$.

(c) $\mathbb{E}\|g\|^4 = \mathbb{E}(\sum_i g_i^2)^2 = \sum_i \mathbb{E}g_i^4 + \sum_{i \neq j} \mathbb{E}g_i^2 \mathbb{E}g_j^2 = 3d + d(d-1) = d^2 + 2d$. Hölder with exponents 3 and $\frac{3}{2}$ applied to $\|g\|^2 = \|g\|^{2/3} \cdot \|g\|^{4/3}$ gives $\mathbb{E}\|g\|^2 \leq (\mathbb{E}\|g\|^{2/3})^{3/2} (\mathbb{E}\|g\|^4)^{1/3}$, so $\mathbb{E}\|g\| \geq (\mathbb{E}\|g\|^2)^{3/2} / (\mathbb{E}\|g\|^4)^{1/2} = d^{3/2} / \sqrt{d^2 + 2d} = \sqrt{d} / \sqrt{1 + 2/d} \geq \sqrt{d}(1 - 1/d) \geq \sqrt{d} - 1$, using $(1+x)^{-1/2} \geq 1 - x/2$. At $d = 100$ the bounds are 9 and 10; the exact value is $\sqrt{2} \Gamma(\frac{101}{2}) / \Gamma(50) = 9.975$.

1.11 A (a) Sum: $\phi(t) = (\phi_1(t), \phi_2(t)) \in \mathbb{R}^{k_1+k_2}$ gives $\langle \phi(s), \phi(t) \rangle = (\Sigma_1)_{s,t} + (\Sigma_2)_{s,t}$. Product: $\phi(t) = (\phi_1(t)_a \phi_2(t)_b)_{a \leq k_1, b \leq k_2} \in \mathbb{R}^{k_1 k_2}$ gives

$$\langle \phi(s), \phi(t) \rangle = \sum_{a,b} \phi_1(s)_a \phi_1(t)_a \phi_2(s)_b \phi_2(t)_b = (\Sigma_1)_{s,t} (\Sigma_2)_{s,t}.$$

A kernel with a feature map is positive semidefinite (Lemma 1.2 applied to the process it defines, or directly: $\sum_{ij} c_i c_j \Sigma_{t_i, t_j} = \|\sum_i c_i \phi(t_i)\|^2 \geq 0$).

(b) $\langle u, v \rangle$ is a kernel on $\mathbb{R}^d$ with $\phi = \text{id}$; by (a), so is every power $\langle u, v \rangle^m$, with feature map $u \mapsto u^{\otimes m} = (u_{a_1} \cdots u_{a_m})_{a_1, \dots, a_m} \in \mathbb{R}^{d^m}$. (So is every polynomial in $\langle u, v \rangle$ with nonnegative coefficients, and in the limit $e^{\langle u, v \rangle}$ — the kernel behind the Gaussian-kernel machines of statistics.)

(c) The feature map for $\langle u, v \rangle^2$ from (a) is $\phi(u) = (u_a u_b)_{a,b} = uu^\top$, a $d \times d$ matrix, with the Frobenius inner product. The process is $X_u = \langle uu^\top, G \rangle = \sum_{a,b} u_a u_b G_{ab} = u^\top G u$ for G a $d \times d$ matrix with i.i.d. $\mathcal{N}(0, 1)$ entries. Check: $\mathbb{E}[u^\top G u v^\top G v] = \sum_{a,b,c,e} u_a u_b v_c v_e \delta_{ac} \delta_{be} = \langle u, v \rangle^2$.

1.12 A (a) $\mathbf{1}_{[0,s]} \mathbf{1}_{[0,t]} = \mathbf{1}_{[0, \min(s,t)]}$, whose integral is $\min(s, t)$.

(b) (B_r, B_s, B_t) is a Gaussian vector with covariance given by $\min$, so $B_t - B_s$ is Gaussian with variance $\Sigma_{t,t} - 2\Sigma_{s,t} + \Sigma_{s,s} = t - 2s + s = t - s$, and $\text{Cov}(B_t - B_s, B_r) = \min(t, r) - \min(s, r) = r - r = 0$ for $r \leq s$. Jointly Gaussian plus uncorrelated is independent, Theorem 1.16 of Lectures 1–2. (Through the feature map: $B_t - B_s = \langle \mathbf{1}_{(s,t]}, g \rangle$, and $\mathbf{1}_{(s,t]} \perp \mathbf{1}_{[0,r]}$.)

1.13 B (a) Theorem 2.9 of Lectures 1–2 with $\Sigma_{t,t} = t$, $\Sigma_{t,1} = \min(t, 1) = t$, $\Sigma_{1,1} = 1$: $\mathbb{E}[B_t | B_1] = \Sigma_{t,1} \Sigma_{1,1}^{-1} B_1 = t B_1$ and $\text{Var}(B_t | B_1) = t - t^2 = t(1 - t)$. So $B_t = t B_1 + (B_t - t B_1)$ with the bridge $B_t - t B_1$ independent of B_1 , Gaussian with variance $t(1 - t)$: zero at both ends, largest in the middle.

(b) Let $t \in [t_j, t_{j+1}]$ (with $t_0 = 0$, $B_0 = 0$) and let $L_t = B_{t_j} + \frac{t - t_j}{t_{j+1} - t_j} (B_{t_{j+1}} - B_{t_j})$ be the linear interpolation. By Theorem 2.9 of Lectures 1–2 the conditional mean is the unique linear combination $\sum_i a_i B_{t_i}$ with $B_t - \sum_i a_i B_{t_i}$ uncorrelated with every B_{t_k} ; L_t is such a combination, so it suffices to check $\text{Cov}(B_t - L_t, B_{t_k}) = 0$ for each k . Write $\theta = \frac{t - t_j}{t_{j+1} - t_j}$, so $B_t - L_t = (1 - \theta)(B_t - B_{t_j}) - \theta(B_{t_{j+1}} - B_{t_j})$. For $k \leq j$, both increments are independent of B_{t_k} by Exercise 1.12(b). For $k \geq j + 1$, $\text{Cov}(B_t - B_{t_j}, B_{t_k}) = t - t_j$ and $\text{Cov}(B_{t_{j+1}} - B_{t_j}, B_{t_k}) = t_{j+1} - t_j$, so the covariance is $(1 - \theta)(t - t_j) - \theta(t_{j+1} - t_j) = 0$ by the choice of θ . (For t beyond t_n the same computation with no right endpoint gives $\mathbb{E}[B_t | \dots] = B_{t_n}$.) The conditional variance is $\theta(1 - \theta)(t_{j+1} - t_j)$ by (a) rescaled, so the posterior of Brownian motion given n points is a chain of independent bridges — exactly the picture one draws.

1.14 C (a) For each g , $\sup_t X_t = \sup_t \langle \phi(t), g \rangle = \sup_{h \in S} \langle h, g \rangle$: the two suprema run over the same set of numbers. Take expectations.

(b) For fixed $h_0 \in S$, $\sup_{h \in S} \langle h, g \rangle \geq \langle h_0, g \rangle$, whose expectation is 0; monotonicity in S is monotonicity of the supremum. $\langle h + x, g \rangle = \langle h, g \rangle + \langle x, g \rangle$ with $\mathbb{E}\langle x, g \rangle = 0$, and $\langle \lambda h, g \rangle = \lambda \langle h, g \rangle$. A linear function attains its supremum over the convex hull at a point of the set: if $h = \sum_j \theta_j h_j$ with $\theta_j \geq 0$, $\sum_j \theta_j = 1$, then $\langle h, g \rangle = \sum_j \theta_j \langle h_j, g \rangle \leq \max_j \langle h_j, g \rangle$, so $w(\text{conv } S) \leq w(S)$, and $\geq$ is monotonicity.

(c) The supremum of $\langle h, g \rangle$ over a unit ball is the dual norm: $\sup_{\|h\|_2 \leq 1} \langle h, g \rangle = \|g\|_2$ (Cauchy–Schwarz, equality at $h = g/\|g\|$), $\sup_{\|h\|_1 \leq 1} \langle h, g \rangle = \max_i |g_i|$ (attained at $h = \pm e_i$), $\sup_{\|h\|_\infty \leq 1} \langle h, g \rangle = \|g\|_1$ (attained at $h = \text{sign}(g)$). Then: $\sqrt{k} - 1 \leq \mathbb{E}\|g\|_2 \leq \sqrt{k}$ is Exercise 1.10(c). $\max_i |g_i| = \max_{j \leq 2k} Y_j$ with $Y = (g_1, \dots, g_k, -g_1, \dots, -g_k)$, $2k$ standard Gaussians, dependent, and Proposition 1.9 does not care: $\leq \sqrt{2 \log(2k)}$. $\mathbb{E}\|g\|_1 = k \mathbb{E}|g_1| = k \sqrt{2/\pi}$, since $\mathbb{E}|g_1| = 2 \int_0^\infty x e^{-x^2/2} dx / \sqrt{2\pi} = \sqrt{2/\pi}$.

(d) For $h \in S_k$ with support I , $|I| \leq k$, Cauchy–Schwarz on the coordinates in I gives $\langle h, g \rangle \leq \|h\|_2 \left(\sum_{i \in I} g_i^2 \right)^{1/2} \leq \left(\sum_{i \in I} g_i^2 \right)^{1/2}$, with equality when h is proportional to $(g_i)_{i \in I}$; maximizing over I picks the k largest $|g_i|$, so $\sup_{h \in S_k} \langle h, g \rangle = \left(\sum_{i \leq k} |g_{(i)}|^2 \right)^{1/2}$ with $|g_{(1)}| \geq |g_{(2)}| \geq \dots$ the order statistics of $|g_i|$. This is at most $\sqrt{k} \max_i |g_i|$, and (c) gives $w(S_k) \leq \sqrt{k} \sqrt{2 \log(2d)}$. Against $w(B_2^d) \geq \sqrt{d} - 1$: the sparse set has width of order $\sqrt{k \log d}$, as if it lived in $k \log d$ dimensions. The number of Gaussian measurements needed to recover a k -sparse vector is, up to constants, the square of its width — $k \log d$ rather than d — which is the content of the sharper bound $w(S_k) \leq \sqrt{2k \log(ed/k)}$ and of Gordon’s escape theorem.

1.15 B (a) $\text{Cov}(X_i, X_j) = \rho + (1 - \rho)\delta_{ij}$: all variances 1, all correlations ρ . Since g_0 does not depend on i , $\max_i X_i = \sqrt{\rho} g_0 + \sqrt{1 - \rho} \max_i g_i$, and $\mathbb{E}g_0 = 0$ gives $\mathbb{E} \max_i X_i = \sqrt{1 - \rho} \mathbb{E} \max_i g_i \approx \sqrt{(1 - \rho) 2 \log n}$. The union bound $\sqrt{2 \log n}$ is attained only at $\rho = 0$; at $\rho = 1$ all X_i coincide and the maximum is a single $\mathcal{N}(0, 1)$ with mean 0.

(b) The union bound (Proposition 1.9, $\text{Var } X_i = \|u_i\|^2 = 1$) gives $\sqrt{2 \log n}$. For the other bound, $\max_i \langle u_i, g \rangle \leq \sup_{\|u\|=1} \langle u, g \rangle = \|g\|$ pointwise, so $\mathbb{E} \max_i X_i \leq \mathbb{E}\|g\| \leq \sqrt{d}$ (Exercise 1.10). The first bound is sharp for $n \leq d$ orthonormal u_i : then the X_i are i.i.d. standard and Proposition 2.3 gives $\sqrt{2 \log n}(1 + o(1))$. The second is sharp when the u_i are spread over the whole sphere: if every unit vector is within ε of some u_i , then the maximizer $u^* = g/\|g\|$ has a u_i with $\|u_i - u^*\| \leq \varepsilon$, so $\max_i X_i \geq \langle u_i, g \rangle - \varepsilon \|g\| = (1 - \varepsilon)\|g\|$ and $\mathbb{E} \max_i X_i \geq (1 - \varepsilon)\mathbb{E}\|g\|$. Lecture 10 counts how many points that takes: $e^{\Theta(d)}$, which is where the two bounds cross.

1.16 B (a) $X_t = \langle \phi(t), g \rangle = 0$ iff $\phi(t) \perp g$ iff $\phi(t) \in g^\perp$; since $g \neq 0$ a.s., $g^\perp$ is a hyperplane through the origin.

For a rotation U , $Ug \sim \mathcal{N}(0, UU^\top) = \mathcal{N}(0, I_k)$ (Theorem 1.5 of Lectures 1–2), and $(Ug)^\perp = U(g^\perp)$, so $U(g^\perp)$ has the same law as $g^\perp$. Equivalently: the unit normal $g/\|g\|$ is uniform on S^{k-1} .

(b) By (a), $\#\{t : X_t = 0\}$ is the number of times the curve ϕ — equivalently its projection γ , since $\phi(t) \in g^\perp$ iff $\gamma(t) \in g^\perp$ — crosses the random hyperplane $g^\perp$. Let $f(A)$ be the expected number of crossings for a curve A on the sphere. It is additive when a curve is cut into pieces (a random hyperplane misses any given finite set of points with probability one), and by (a) it is invariant under rotations. All arcs of great circles of length ℓ are rotations of one another, so f of such an arc is a function $F(\ell)$ of the length alone; it is additive, $F(\ell + \ell') = F(\ell) + F(\ell')$, and nondecreasing, hence $F(\ell) = c\ell$. A full great circle C , of length 2π , is the unit circle of a 2-plane P ; the hyperplane $g^\perp$ meets P in a line through the origin (with probability one it does not contain P), and a line through the center meets a circle in exactly two points. So $2 = F(2\pi) = 2\pi c$ and $c = 1/\pi$. For a smooth γ , inscribe great-circle polygons: their lengths tend to $\text{length}(\gamma)$ and their crossing counts to those of γ , giving $\mathbb{E}\# = \text{length}(\gamma)/\pi$. This is Crofton's formula; in the plane, the same argument is Buffon's needle.

(c) $X_\theta = \langle \phi(\theta), g \rangle$ with $\phi(\theta) = (\cos \theta, \sin \theta, \cos 2\theta, \sin 2\theta, \dots, \cos n\theta, \sin n\theta) \in \mathbb{R}^{2n}$ and $g = (a_1, b_1, \dots, a_n, b_n)$. Then $\|\phi(\theta)\|^2 = \sum_{j \leq n} (\cos^2 j\theta + \sin^2 j\theta) = n$, constant, so $\gamma = \phi/\sqrt{n}$ and $\|\gamma'(\theta)\|^2 = \frac{1}{n} \sum_{j \leq n} j^2 = \frac{(n+1)(2n+1)}{6}$. By (b),

$$\mathbb{E}\#\{\theta \in [0, 2\pi) : X_\theta = 0\} = \frac{1}{\pi} \cdot 2\pi \sqrt{\frac{(n+1)(2n+1)}{6}} = 2\sqrt{\frac{(n+1)(2n+1)}{6}} \sim \frac{2n}{\sqrt{3}}.$$

The feature curve winds around a sphere of radius $\sqrt{n}$ in $\mathbb{R}^{2n}$ at constant speed, and a random hyperplane cuts it about $2n/\sqrt{3}$ times — a bit more than the 2 cuts of the great circle $n = 1$ per unit of frequency, because higher harmonics make the curve longer. (For comparison, the real zeros of a random *algebraic* polynomial of degree n with i.i.d. Gaussian coefficients number only $\sim \frac{2}{\pi} \log n$: the feature curve $(1, x, \dots, x^n)$, projected to the sphere, is short.)

§2 Suprema

2.14 A (a) Put $t_j = \min\{(2j-1)\varepsilon^2, 1\}$ for $j = 1, \dots, \lceil 1/(2\varepsilon^2) \rceil$. Every $t \in [0, 1]$ is within ε^2 of some t_j in $|\cdot|$, hence within ε in $d = \sqrt{|\cdot|}$, so $N([0, 1], d, \varepsilon) \leq \lceil 1/(2\varepsilon^2) \rceil \leq 1 + 1/(2\varepsilon^2)$. The diameter is 1, and Theorem 2.9 gives $\mathbb{E} \sup_{t \leq 1} B_t \leq 12 \int_0^{1/2} \sqrt{\log(1 + 1/(2\varepsilon^2))} d\varepsilon = 12 \times 0.806 = 9.68$, finite. The exact value, $\sqrt{2/\pi}$, comes from the reflection principle, $\sup_{t \leq 1} B_t \stackrel{d}{=} |B_1|$ — a statement about paths, outside this course.

(b) Points $t_j = \min\{(2j-1)\varepsilon/L, 1\}$, $j \leq \lceil L/(2\varepsilon) \rceil$, are within ε/L of every t , hence within ε in d ; so $N \leq 1 + L/(2\varepsilon)$. Also $d(s, t) \leq \sqrt{\text{Var } X_s} + \sqrt{\text{Var } X_t} \leq 2$, so the integral in Theorem 2.9 runs over at most $[0, 1]$. For $\varepsilon \leq 1$, $1 + L/(2\varepsilon) \leq (1 + L)/\varepsilon$, so with $A = 1 + L$ the given estimate yields $\mathbb{E} \sup X_t \leq 12(\sqrt{\log(1 + L)} + 1)$. (The estimate itself: $\sqrt{\log(A/\varepsilon)} \leq \sqrt{\log A} + \sqrt{\log(1/\varepsilon)}$ and $\int_0^1 \sqrt{\log(1/\varepsilon)} d\varepsilon = \Gamma(\frac{3}{2}) < 1$.) The order $\sqrt{\log L}$ is that of the maximum of L independent standard Gaussians: a process that changes on scale $1/L$ has about L genuinely different values on $[0, 1]$. Check: the kernel $\Sigma_{s,t} = e^{-L^2(s-t)^2/2}$ has $d(s, t)^2 = 2(1 - e^{-L^2(s-t)^2/2}) \leq L^2(s-t)^2$; at $L = 50$ a simulation gives $\mathbb{E} \sup = 2.27$, against $\sqrt{2 \log L} = 2.80$ and the bound 34.

(c) That nearby points carry nearly the same value: the union bound treats $B_{j/n}$ and $B_{(j+1)/n}$ as two unrelated Gaussians, while chaining bounds their *difference*, whose standard deviation $1/\sqrt{n}$ is small, and pays $\sqrt{\log(\text{number of links})}$ only against that small scale.

2.15 B (a) Let $1 - \varepsilon \leq \|x\| \leq 1$, so $x \neq 0$, and put $u = x/\|x\| \in S^{d-1}$. Some u_i has $\|u - u_i\| \leq \varepsilon$, and $\|x - u\| = 1 - \|x\| \leq \varepsilon$, so $\|x - u_i\| \leq 2\varepsilon$. Hence the n balls $B(u_i, 2\varepsilon)$ cover the shell $\{1 - \varepsilon \leq \|x\| \leq 1\}$, whose volume is $(1 - (1 - \varepsilon)^d) \text{vol}(B_2^d)$; the balls have total volume $n(2\varepsilon)^d \text{vol}(B_2^d)$. So $n(2\varepsilon)^d \geq 1 - (1 - \varepsilon)^d$.

(b) At $\varepsilon = \frac{1}{4}$: $n \geq 2^d(1 - (3/4)^d) \geq 2^d(1 - \frac{9}{16}) = \frac{7}{16}2^d \geq 2^{d-2}$ for $d \geq 2$. With Lemma 2.6, $2^{d-2} \leq N(S^{d-1}, \|\cdot\|, \frac{1}{4}) \leq 9^d$: the covering number of the sphere at a fixed scale is $e^{\Theta(d)}$.

(c) Sudakov's inequality at $\varepsilon = \frac{1}{4}$ with (b): $\mathbb{E}\|g\| = \mathbb{E} \sup_u \langle g, u \rangle \geq c \cdot \frac{1}{4} \sqrt{\log 2^{d-2}} = \frac{c}{4} \sqrt{(d-2) \log 2}$, which is $c'\sqrt{d}$ for $d \geq 4$, say. Example 2.11 gives $\mathbb{E}\|g\| \leq 12c_0\sqrt{d}$ from the upper bound $N \leq (1 + 2/\varepsilon)^d$. So $\mathbb{E}\|g\| \asymp \sqrt{d}$ follows from $N(S^{d-1}, \|\cdot\|, \varepsilon) = e^{\Theta(d)}$ at a fixed scale, and from nothing else about the sphere. (Of course Exercise 1.10 gives the sharp $[\sqrt{d} - 1, \sqrt{d}]$ directly; the point is that the covering numbers alone already know the order.)

2.16 B (a) $Y_{(u,v)} = \sum_{ij} v_i u_j M_{ij} = \langle vu^\top, M \rangle_F$ is linear in the entries of M , and for a fixed matrix A , $\text{Var}\langle A, M \rangle_F = \sum_{ij} A_{ij}^2 = \|A\|_F^2$; so $d((u, v), (u', v'))^2 = \text{Var}(\langle vu^\top - v'(u')^\top, M \rangle_F) = \|vu^\top - v'(u')^\top\|_F^2$. Expanding, with $\|ab^\top\|_F = \|a\|\|b\|$ and $\langle ab^\top, cd^\top \rangle_F = \langle a, c \rangle \langle b, d \rangle$, this is $1 + 1 - 2\langle v, v' \rangle \langle u, u' \rangle$. For the triangle bound write $vu^\top - v'(u')^\top = v(u - u')^\top + (v - v')(u')^\top$, so $d \leq \|v\|\|u - u'\| + \|v - v'\|\|u'\| = \|u - u'\| + \|v - v'\|$.

(b) Let U be an $\varepsilon/2$ -net of S^{d-1} and V an $\varepsilon/2$ -net of S^{N-1} , of the sizes in Lemma 2.6. For $(u, v) \in T$ pick $u' \in U$, $v' \in V$ within $\varepsilon/2$ of u, v ; by (a), $d((u, v), (u', v')) \leq \varepsilon$. So $U \times V$ is an ε -net, of size at most $(1 + 4/\varepsilon)^d (1 + 4/\varepsilon)^N$.

And $D \leq \|vu^\top\|_F + \|v'(u')^\top\|_F = 2$. Theorem 2.9:

$$\mathbb{E}\|M\|_{\text{op}} = \mathbb{E} \sup_T Y \leq 12 \int_0^1 \sqrt{(N+d) \log(1+4/\varepsilon)} d\varepsilon = 12c_1 \sqrt{N+d}, \quad c_1 = 1.559,$$

and $12c_1 = 18.7 \leq 19$, $\sqrt{N+d} \leq \sqrt{N} + \sqrt{d}$. For $N = 100$, $d = 50$ the bound is 229 against the truth 16.6.

(c) By (a) and $|\langle v, v' \rangle| \leq 1$, $d^2 \geq 2 - 2|\langle u, u' \rangle| = \min(\|u - u'\|^2, \|u + u'\|^2)$. Fix any $u \in S^{d-1}$ and any v ; some net point (u_i, v_i) is within ε of (u, v) , so u is within ε of u_i or of $-u_i$. Hence the $2n$ points $\pm u_i$ form an ε -net of S^{d-1} , $2n \geq N(S^{d-1}, \varepsilon)$, and by symmetry $2n \geq N(S^{N-1}, \varepsilon)$. At $\varepsilon = \frac{1}{4}$, Exercise 2.15(b) gives $n \geq \frac{1}{2} 2^{\max(N, d)-2} = 2^{\max(N, d)-3}$.

(d) Sudakov's inequality at $\varepsilon = \frac{1}{4}$: $\mathbb{E}\|M\|_{\text{op}} \geq \frac{c}{4} \sqrt{(\max(N, d) - 3) \log 2} \geq c' \sqrt{\max(N, d)} \geq \frac{c'}{2} (\sqrt{N} + \sqrt{d})$ for $\max(N, d) \geq 6$, say (and $\mathbb{E}\|M\|_{\text{op}} \geq \mathbb{E}|M_{11}| > 0$ covers the rest). With (b), $c''(\sqrt{N} + \sqrt{d}) \leq \mathbb{E}\|M\|_{\text{op}} \leq 19(\sqrt{N} + \sqrt{d})$: the order of the operator norm of a Gaussian matrix follows from the covering numbers of $S^{N-1} \times S^{d-1}$ alone. Lecture 11 gives the constant 1 in the upper bound by comparison.